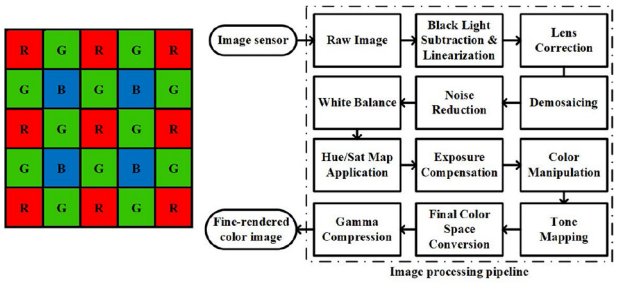
Histogram of Oriented Gradients Feature Extraction From Raw Bayer Pattern Images

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***Abstract*—This brief studies the redundancy in the image processing pipeline for histogram of oriented gradients (HOG) feature extraction. The impact of demosaicing on the extracted HOG features is analyzed and experimented. It is shown that by taking advantage of the inter-channel correlation of natural images, the HOG features can be directly extracted from the Bayer pattern images with proper gamma compression. Due to the elimination of the image processing pipeline, the power consumption and computational complexity of the detection system can be significantly reduced. Experimental results show that the Bayer pattern image-based HOG features can be used in pedestrian detection systems with little performance degradation.**

***Index Terms*—Histogram of oriented gradients (HOG), Bayer pattern, object detection.**



(a) (b)

Fig. 1. (a) The RGGB Bayer color filter array (CFA) pattern. (b) The image processing pipeline in the Adobe DNG converter.

I. Introduction

OBJECT detection is one of the fundamental problems in computer vision. Due to the redundancy in images, the most important information, i.e., the features, need to be extracted before detection. In the past decades, numerous feature descriptors have been proposed. Histogram of oriented gradients (HOG) is one of the well-recognized features due to its superior performance and relatively simple computation. Since first proposed in [1], HOG and its derivatives have been widely used in various pedestrian and object detection algorithms [2]–[4]. Even with the popularity of convolutional neuron network (CNN), HOG-based algorithms are still favorable in many applications due to their balanced tradeoff between accuracy and complexity.

Digital images are the raw inputs for computer vision algorithms, including HOG-based object detection systems. It is well-known that three channels, i.e., Red (R), Green (G) and Blue (B), are needed to produce a color image. For most of the modern digital cameras and smart phones, images are captured by a monochromatic image sensor covered by a Bayer color filter array (CFA) [5] as shown in Fig. 1(a). The primary output from the sensor is therefore a mosaiced single-channel Bayer pattern image. This Bayer pattern image is converted to a three-channel color image by interpolating the other two missing color components. This process, which consumes a significant amount of processing time and power [6], is always referred to as demosaicing. There are many other steps in the image processing pipeline, which may vary from one manufacture to another. Shown in Fig. 1(b) is a 11-step image processing pipeline from the Adobe DNG converter [7].

In order to increase the processing speed and to reduce the power consumption, a lot of research effort has been devoted to optimize the computation of HOG features [8]–[10]. However, in the field of computer vision, the origins of the images are always neglected by researchers and the redundancy in the image processing pipeline has seldom been explored. Almost all existing HOG-based algorithms take the finely rendered color images (after image processing pipeline) as inputs. But it is well-known that the existing image processing pipeline is designed and tuned for optimal perceptual image quality perceived by human eyes. However, pleasing reproductions of the scenes does not necessarily translates into optimal performance of computer vision applications. Moreover, it has been shown that cumulative errors may be introduced by the image processing pipeline, which may harm or even damage the original information from the image sensor [11]. Therefore, for HOG-based object detection algorithms, the configuration of the image processing pipeline or even its necessity needs to be reconsidered.

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In a recent paper, Buckler *et al.* use an empirical approach to study the image processing pipeline's impact on the computer vision performances [6]. Extensive experiments based on eight existing vision algorithms are conducted, and a minimal image processing pipeline consisting of denoise, demosaicing and gamma compression is proposed. But the minimal configuration of the image processing pipeline for HOG is not addressed in [6]. There are also some studies that try to completely bypass the traditional image processing pipeline and extract the high-level features such as local binary pattern (LBP) from Bayer pattern images [12]–[16]. In [16], gradient operators designed specially for Bayer pattern images are proposed. Filter structures are mapped to the color filter arrangement of Bayer pattern subsampling, where only the green channel is used to extract the HOG features. In some hardware implementation of HOG-based detection algorithms, the redundancy in the image processing pipeline has been explored to reduce the hardware complexity [17]–[19]. They extract features from raw images by applying a down-sampling operation (e.g., weighted average four pixels in each Bayer structure to produce one pixel) to "demosaic" the Bayer pattern images. However, due to the lack of theoretically analysis and experiments, the influence of the image processing pipeline on HOG-based algorithms is still unclear.

In this brief, the impact of the demosaicing step as well as other common steps in the conventional image processing pipeline on HOG feature extraction is studied and experimented. Instead of demosaicing the Bayer pattern images, it is found that by taking advantage of the inter-channel correlation of natural images, the HOG features can be directly extracted from the Bayer pattern images, where proper gamma compression may be needed. It is shown in the experiments that the HOG features based on Bayer pattern images can be used in pedestrian detection algorithms with little performance degradation.

The rest of this brief is organized as follows. Section II discusses the backgrounds, including the image processing pipeline and HOG features. Section III presents the proposed HOG feature extraction from the Bayer pattern images. Experimental results are presented in Section IV and conclusions are drawn in Section V.

II. Background

A. The Image Processing Pipeline

The image processing pipeline for photography consists of a series of processing steps to convert the raw image signal acquired from an image sensor to a finely rendered color image. It is well-known that for different manufactures, the image processing pipeline may vary in the design of algorithm for each step and the order of processing steps. This section mainly focus on the steps in the minimal pipeline shown in Fig. 2(a), which has been proposed in [6].

- Demosaicing: Demosaicing is a crucial step to convert a single-channel Bayer pattern image to a three-channel color image by interpolating the other two missing color components at each pixel. In order to minimize the color artifacts, sophisticated demosaicing algorithms usually consume a lot of computation resources and processing time [6].
- Tone mapping: This step is to compress the image's dynamic range to fit the display requirements (normally 8-bit per pixel) while maintaining a pleasing visual contrast. Gamma compression is one of the most commonly used global tone mapping methods. It transforms the intensity of a pixel $I^k(x, y)$ to $I'^k(x, y)$ by

$$I'^k(x, y) = A \times \left(I^k(x, y)\right)^{\gamma}, \tag{1}$$

  where x and y represent the x-coordinate and y-coordinate of a pixel location, respectively, k represents one of the three channels among R, G and B. A and γ are the two transformation parameters to determine the effect of gamma compression.
- Denoise: Denoise is used to suppress the noises introduced by the image sensor itself and by the imaging process steps.

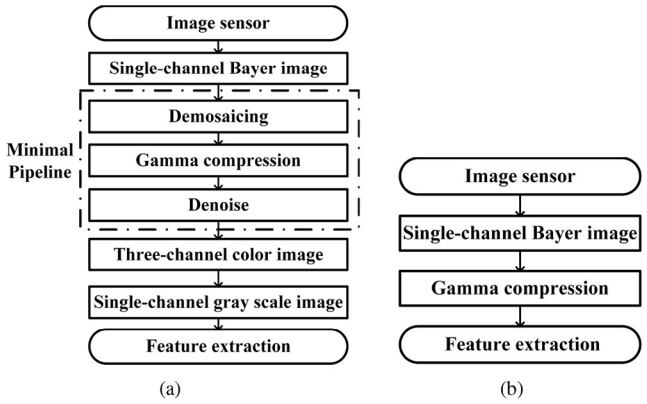


Fig. 2. Feature extraction process with (a) the minimal pipeline [6] and (b) the proposed pipeline.

B. The HOG Descriptor

HOG is a feature descriptor initially proposed for pedestrian detection [1]. It counts the number of occurrences of gradient orientation in a detection window. The main steps to compute HOG features can be summarized as follows.

1) Gradient computation: In this step, the spatial gradients in horizontal and vertical directions are computed. These two gradients are then used to compute the gradient magnitudes and angles.
2) Orientation binning: In this step, the image is divided into small connected regions called cells. The gradient magnitude of each pixel in a cell is voted into different orientation bins according to the gradient angle.
3) Feature description: In this step, adjacent cells are grouped into blocks. Each block is normalized by its L_2-norm. The normalized block histograms in a detection window are concatenated to form a descriptor.

There are many derivatives of the original HOG descriptor such as F-HOG [4] and pyramid HOG (p-HOG) [20]. The basic steps of these derivatives are similar.

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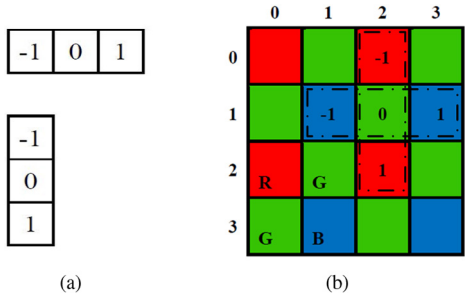


Fig. 3. Gradient computation based on Bayer pattern images. (a) Central difference 1-D masks. (b) Apply the 1-D masks to a Bayer pattern image.

III. Gradient Extraction from Bayer Pattern Images

Image gradient measures the change of intensity. Mathematically, for a two-dimensional function $f(x, y)$, the gradients can be computed by the derivatives with respect to x and y. For a digital image where x and y are discrete values, the derivatives can be approximated by finite differences [21].

One of the most commonly used approaches for image gradient computation is the central difference based method. The gradient at pixel location (x, y) can be computed as

$$G_x = I(x+1, y) - I(x-1, y), \tag{2}$$
$$G_y = I(x, y+1) - I(x, y-1). \tag{3}$$

Here G_x and G_y represent the gradients in the horizontal and vertical directions respectively. Note that the computation of (2) and (3) can be implemented by convolving images with two 1-D masks of $[-1, 0, 1]$ and $[-1, 0, 1]^T$ as shown in Fig. 3(a).

The fundamental idea of this brief is illustrated in Fig. 2(b). Instead of demosaicing the Bayer pattern images before computing the gradients as shown in Fig. 2(a), it is found that by taking advantage of the inter-channel correlation of natural images [22], the gradients can be directly extracted from Bayer pattern images as shown in Fig. 2(b). As it will be shown later, proper gamma compression is needed to adjust the contrast.

Fig. 3 illustrates the gradient computation from Bayer pattern images. For example, applying the 1-D masks presented in Fig. 3(a) at pixel location $(1, 2)$ generates

$$G^B_{x(1,2)} = I^B_{(1,3)} - I^B_{(1,1)}, \tag{4}$$
$$G^R_{y(1,2)} = I^R_{(2,2)} - I^R_{(0,2)}, \tag{5}$$

where G^B and G^R are the gradients of the blue and red channels, respectively. As shown in Fig. 3, the two input pixels for coefficients 1 and -1 in the convolution masks are from the same channel, i.e., differences are computed on homogeneous pixels. This is true regardless of the pixel locations, as long as the alternating pattern of Bayer images holds. Moreover, according to the inter-channel correlation of natural images [22], we have

$$I^B_{(1,3)} - I^B_{(1,1)} \approx I^{\widehat{R}}_{(1,3)} - I^{\widehat{R}}_{(1,1)} \approx I^{\widehat{G}}_{(1,3)} - I^{\widehat{G}}_{(1,1)}, \tag{6}$$
$$I^R_{(2,2)} - I^R_{(0,2)} \approx I^{\widehat{B}}_{(2,2)} - I^{\widehat{B}}_{(0,2)} \approx I^{\widehat{G}}_{(2,2)} - I^{\widehat{G}}_{(0,2)}. \tag{7}$$

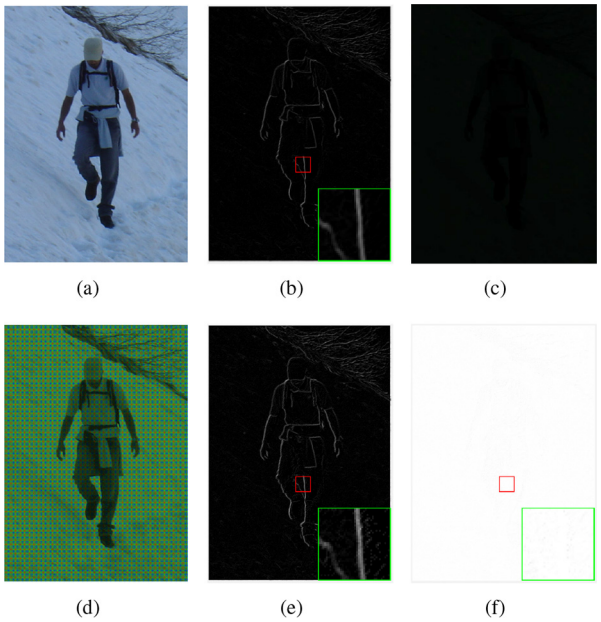


Fig. 4. Comparison of gradients extracted from a color image and its Bayer version. (a) An image from the INRIA pedestrian dataset. (b) Gradient magnitude map generated using (a). (c) The converted Bayer version of (a) using the reverse pipeline in [6]. (d) The Bayer version image after gamma compression. (e) Gradient magnitude map generated using (d). (f) The difference map between (b) and (e).

Here $I^{\widehat{R}}$, $I^{\widehat{G}}$ and $I^{\widehat{B}}$ are the missing color components at the corresponding pixels. Combining (4) and (5) with (6) and (7), we have

$$G \approx G^G \approx G^R \approx G^B, \tag{8}$$

which means the gradients can be computed using the information of a single channel. According to (4), (5) and (8), it can be concluded that gradients can be directly extracted from the Bayer pattern images without demosaicing, as long as the inter-channel correlation holds. To validate the above statement, we extract the gradient maps from a color image as well as its corresponding Bayer version (after gamma compression). These two gradient maps as well as the difference map are illustrated in Fig. 4. As can be observed from Fig. 4(b) and Fig. 4(e), the gradient maps look very close to each other. A closer look (e.g., in the zoomed area) reveals that Fig. 4(e) contains more noise in the whole map than that in Fig. 4(b). The difference map shown in Fig. 4(f) is almost all white, meaning that the two gradient maps are almost the same at all pixel locations. This means the noise level in Fig. 4(e) is rather low.

IV. Experiments

In this section, experimental results are presented to validate the proposed Bayer pattern image-based HOG feature extraction pipeline. The INRIA dataset [23], which is a popular dataset for pedestrian detection, is used in this brief. In our experiments, all the Bayer version images are generated by reversing the original color images using the reverse pipeline proposed in [6].

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TABLE I
COMPARISON OF DIFFERENT GRADIENT MAPS

Datasets	Gradient maps	Averaged results		
		MSSIM	MSE	PSNR
Converted Bayer images	X Direction	0.635	154.629	26.979
	Y Direction	0.617	155.431	26.910
	Magnitude	0.482	376.848	23.086
Gamma Compression only	X Direction	0.865	28.776	33.934
	Y Direction	0.863	29.502	33.804
	Magnitude	0.817	66.624	30.288
Gamma Compression+Denoise	X Direction	0.898	24.424	34.756
	Y Direction	0.894	25.075	34.619
	Magnitude	0.855	57.623	31.013

For gamma compression, $A = 2$ and $\gamma = 0.5$ are used.

A. Gradient Map Comparison

In this experiment, the gradient maps generated from the original INRIA dataset and its Bayer version are compared. The evaluation metrics are introduced first followed the comparison results.

To estimate the differences between gradient maps, three commonly used image quality assessment methods, i.e., mean-square error (MSE), peak signal-to-noise ratio (PSNR) and mean structural similarity (MSSIM) are used. Note that here we use SSIM to compare the differences between gradient maps because image gradients measure the change of intensity (luminance), which is a kind of low-level feature that is closely related to the luminance. Listed in Table I are the comparison results of the gradient maps computed from the original color images and their corresponding Bayer versions. Generally speaking, all the three evaluation criteria reveal the same trends. As shown in Table I, according to the three evaluation metrics, the gradient maps extracted from Bayer pattern images with gamma compression are close to that extracted from the color images. This is because as shown in Section III, in the proposed gradient extraction from Bayer pattern images, the inter-channel correction of natural images are used in the same way as that in many demosacing algorithms [22]. It is also noted that the impact of denoising on gradient maps is not significant, while the influence of gamma compression is obvious. This can be visualized in Fig. 5, where three versions of HOG descriptors, i.e., HOG from the original image, HOG from the Bayer pattern image without gamma compression and HOG from the Bayer pattern image with gamma compression, are presented. As shown in Fig. 5(e), the descriptors cannot find enough features in low contrast Bayer pattern image without gamma compression (in Fig. 5(b)). But after adjusting the contrast by gamma compression, the HOG feature extracted from Fig. 5(c) becomes more stable, and close to the one extracted from the original color image in Fig. 5(a).

B. Pedestrain Detection

In this section, we evaluate the pedestrian detection based on HOG features from color images and their corresponding Bayer versions. The traditional frameworks of HOG+support vector machine (SVM) model from [1] and p-HOG+SVM from [24] are used to detect pedestrians from the original INRIA dataset and its Bayer vision. Experimental results are presented using the per-image criteria [2].

As shown in Fig. 6, four different configurations of image processing pipelines are tested in two pedestrain detection frameworks. In Fig. 6, "Full pipeline" means images are generated using a full image processing pipeline as shown

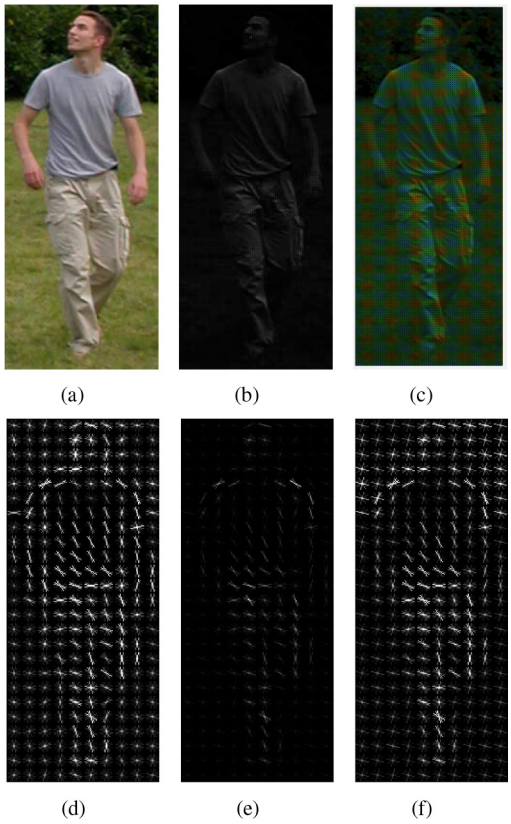


Fig. 5. Comparison of HOG features. (a) An image from the INRIA pedestrian dataset. (b) The converted Bayer version of (a) using the reverse pipeline in [6]. (c) The Bayer version image after gamma compression. (d)-(f): Visualization of the generated HOG descriptors.

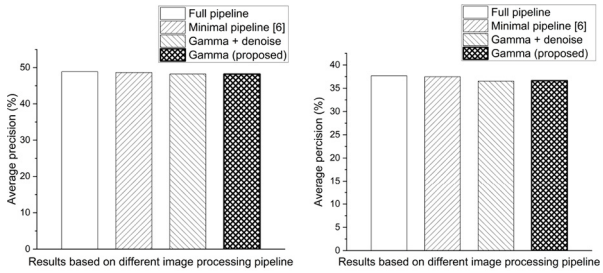


Fig. 6. Comparison of pedestrain detection performance based on HOG features generated from different versions of images using (a) the HOG+SVM framework and (b) the p-HOG+SVM framwork. Full pipeline: images are generated using a full image processing pipeline as shown in Fig. 1(b); Minimal pipeline: images are generated using a minimal image processing pipeline proposed in [6]; Gamma+denoise: images are generated from raw Bayer pattern images by applying gamma compression and denoise; Gamma only: images are generated from raw Bayer pattern images by applying only gamma compression.

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in Fig. 1(b), "Minimal pipeline" means images are generated using the minimal image processing pipeline proposed in [6], "Gamma+denoise" means images are generated from raw Bayer pattern images after applying gamma compression and denoise and "Gamma only" means images are generated from raw Bayer pattern images after applying gamma compression. The experimental results show that no matter in which framework, the average detection rates using different versions of images are very close to each other. Specifically, the differences between different configurations of image processing pipelines are less than 1%. This is because in the (p-)HOG+SVM framework, the only information used for detection is the HOG feature. As shown in Section III and Section IV-A, the HOG features generated from different versions of images, including Bayer pattern images (after gamma compression), are very close to each other. Therefore, the Bayer pattern images can be directly used in HOG feature-based pedestrian detection frameworks. In addition, as in the previous experiment, the influence of denoise is very small, which means the noise observed in Fig. 4(e) is not severe enough to affect the final detection performance. Therefore, it can be concluded that for HOG-based pedestrian detection systems, the complete image processing pipeline except the gamma compression can be skipped with little performance degradation.

As reported in [6], the image processing pipeline consumes a significant portion of power in a computer vision system consisting of an image sensor, an image signal processor and an object detector. More specifically, typical image sensors consume power from 137.1 mW to 338.6 mW and a typical image signal processor dissipates 250mW for processing 1080P video at 60 frame per second (FPS) [6], while a sophisticated object detection chip (based on deformable parts models (DPM), an improved version of HOG) consumes only 58.6 mW for processing 1080P video at 30FPS [25]. From the above statistics, it can be concluded that by skipping almost all the image processing steps as in this brief, the power consumption and the computational complexity of an object detection system can be significantly reduced.

V. Conclusion

In this brief, the necessity of the image processing pipeline, especially the demosaicing step, for HOG-based pedestrian detection system is studied. It is shown that by taking advantage of the inter-channel correlation of natural images, the gradients can be extracted directly from the Bayer pattern images, while proper gamma compression is needed. It is shown that the Bayer pattern image-based HOG features can be used in pedestrian detection frameworks with little performance degradation. Therefore, for HOG-based vision system, the complete image processing pipeline except the gamma compression can be skipped such that the power consumption and computational complexity of the whole detection system can be significantly reduced.

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